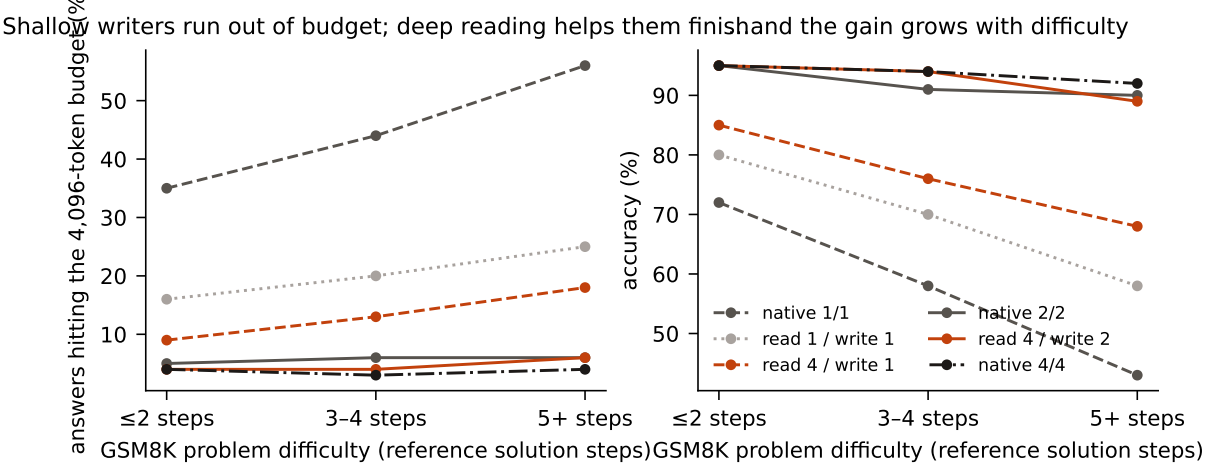
Shallow writers run out of budget; deep reading helps them finishand the gain grows with difficulty
answers hitting the 4,096-token budget (%)
10
20
30
40
50
≤2 steps
3–4 steps
5+ steps
GSM8K problem difficulty (reference solution steps)
accuracy (%)
50
60
70
80
90
native 1/1
read 1 / write 1
read 4 / write 1
native 2/2
read 4 / write 2
native 4/4
≤2 steps
3–4 steps
5+ steps
GSM8K problem difficulty (reference solution steps)